

HW16: Unix Basics

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Using RStudio locally to read my output

Import and read data that was downloaded from remote computer from previous lab.

```
# Read tsv file
b <- read.table(file="results.tsv")
# Set the colnames
colnames(b) <- c("qseqid", "sseqid", "pident", "length", "mismatch", "gapopen", "qstart", "qend", "sstart", "send", "eval", "bitscore")
head(b)
```

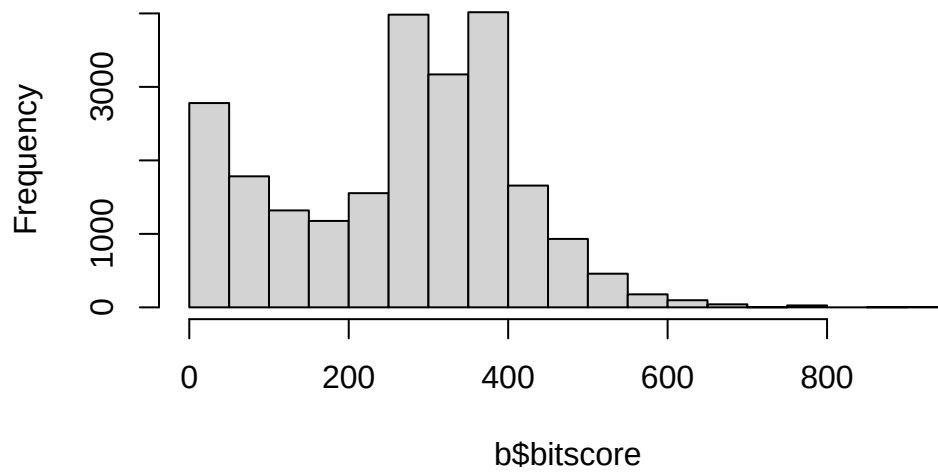
	qseqid	sseqid	pident	length	mismatch	gapopen	qstart	qend	sstart
1	NP_598866.1	XP_073805285.1	46.154	273	130	6	4	267	439
2	NP_598866.1	XP_009294521.1	46.154	273	130	6	4	267	420
3	NP_598866.1	NP_001313634.1	46.154	273	130	6	4	267	476
4	NP_598866.1	XP_009294513.1	46.154	273	130	6	4	267	475
5	NP_598866.1	XP_073797936.1	31.818	132	81	5	4	131	297
6	NP_598866.1	NP_001186666.1	33.071	127	76	5	4	126	338

	send	eval	bitscore
1	703	1.06e-63	216.0
2	684	2.42e-63	214.0
3	740	6.45e-63	214.0
4	739	6.71e-63	214.0
5	423	4.67e-12	68.2
6	459	7.41e-12	67.8

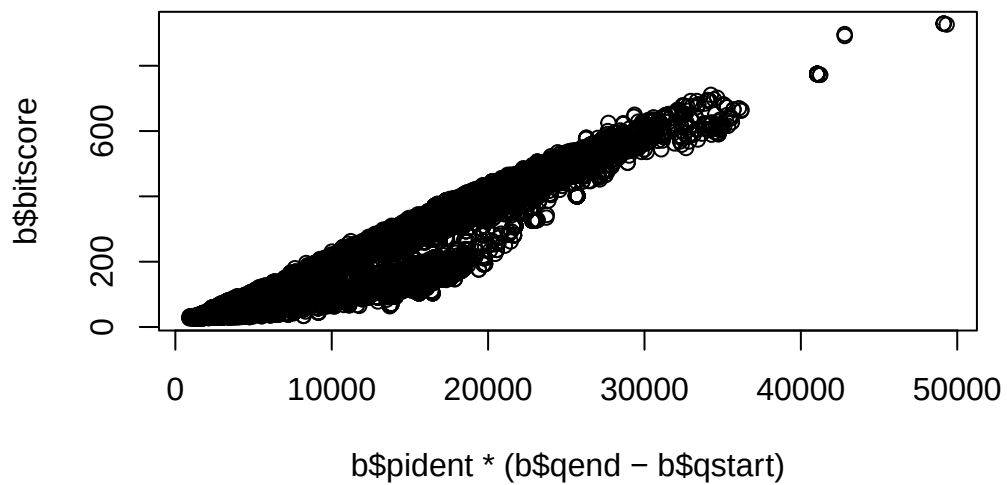
Plot a histogram of bitscore value

```
hist(b$bitscore, breaks = 30)
```

Histogram of b\$bitscore



```
# Blast results are stored in an object called 'b'  
plot(b$pident * (b$qend - b$qstart), b$bitscore)
```

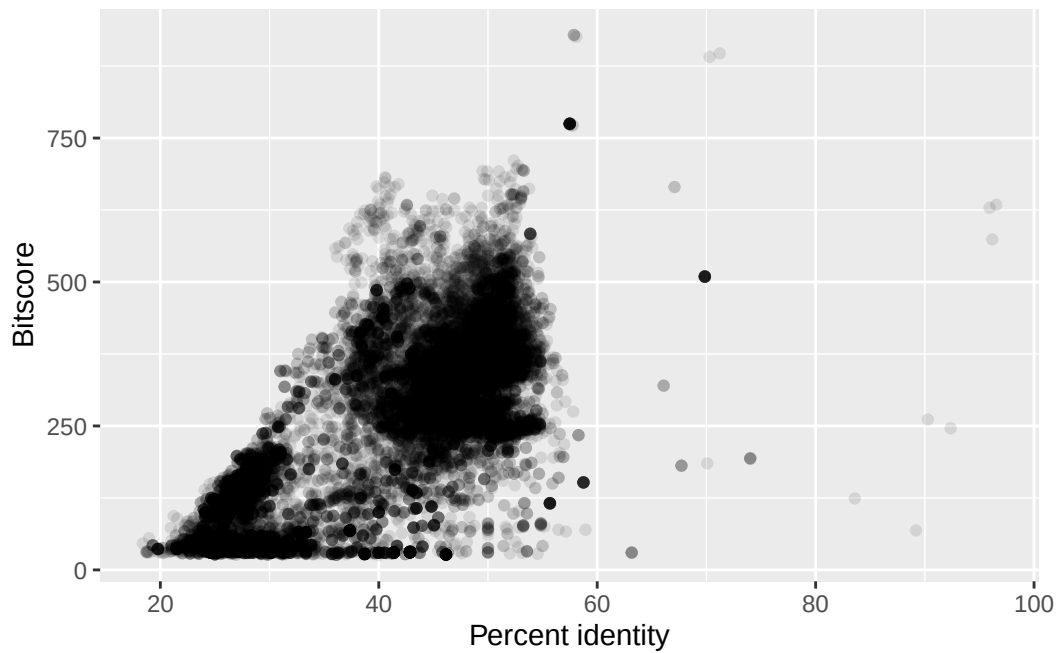


Result visualization

Use **ggplot2** package to plot the values of percent identity and bitscore value.

```
library(ggplot2)

ggplot(b) +
  aes(pident, bitscore) +
  geom_point(alpha=0.1) +
  labs(x = "Percent identity", y = "Bitscore")
```



```
ggplot(b) +
  aes(pident * (qend - qstart), bitscore) +
  geom_point(alpha=0.1) +
  geom_smooth() +
  labs(x = "Percent identity", y = "Bitscore")
```

``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

